

Paper 04 — Agent Memory System

OpenClaw Memory: Identity, Persistence, Evolution

WHYCREMISI RESEARCH PAPERS — N.04
Agent AI Memory System

"It does not reset. It remembers. It grows."

1. Why Memory Is Fundamental

An assistant without memory restarts from zero every time. Memory transforms a tool into a **collaborator**.

WITHOUT MEMORY: Session 100 = Session 1. Generic answers. No growth.

WITH OPENCLAW: "I see you're working on something similar to two weeks ago — would you like to use the same processing chain?"

2. OpenClaw Memory Architecture

LAYER 4	Identity (permanent)	— Who the agent is. Core values. Immutable.
LAYER 3	Personality (slow)	— Communication style. Evolves gradually.
LAYER 2	User Profile (medium)	— Who the user is. Preferences. Plugins.
LAYER 1	Session Memory (daily)	— What happened today. Actions, results.
LAYER 0	Real-Time Context	— BPM, meters, active plugins. Volatile.

3. User Profile Structure

```
{
  "musical_style": { "genres": ["house","techno"], "bpm_range": [120,140] },
  "plugin_preferences": {
    "eq": "FabFilter Pro-Q3", "compressor": "FabFilter Pro-C2",
    "limiter": "FabFilter Pro-L2"
  },
  "processing_preferences": {
    "kick_style": "dry_tight", "master_lufs_target": -14.0,
    "master_peak_ceiling": -0.3, "compression_style": "transparent"
  },
  "acceptance_rates": {
    "eq_suggestions": 0.78, "compression_suggestions": 0.65
  }
}
```

4. Learning Cycle

User accepts EQ suggestion → increment eq_suggestions acceptance_rate
User rejects compression → lower default compression depth for user
User manually corrects freq → update eq_frequency_bias preference
End of session → distil patterns → update user_profile.json

5. Context Injection in Every AI Call

[SESSION CONTEXT]	Date · BPM · Project name
[LIVE AUDIO STATE]	LUFS · Peak · Stereo width · FFT anomaly
[USER PROFILE]	Genre · LUFS target · Preferred plugins
[RELEVANT MEMORY]	Last 3 sessions relevant actions
[ACTIVE PLUGINS]	Per-track plugin chain
[AGENT PERSONALITY]	Style · Language · Verbosity · Technicality

6. Privacy

All data stored locally at `~/.whycremisi/`. Nothing sent to cloud without explicit opt-in. Full `/forget` and `/reset` commands available. API keys never logged in plain text.

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